

§6-2 Volumes + §6-3 Volumes by Cylindrical Shells

Homework(§6-2) : 7,11,18,19,21,22,28,35,41,47,49,50,56,64,65,67

Homework(§6-3) : 3,13,17,20,22,23,29,30,39,41

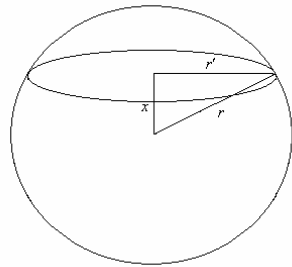
公式：

$$V_3 = \int V_2 dx$$

V_2 (2維截面) = $\left\{ \begin{array}{l} \text{§6.2 (截面為環狀)} \\ \text{§6.3 (截面為圓柱狀)} \end{array} \right.$

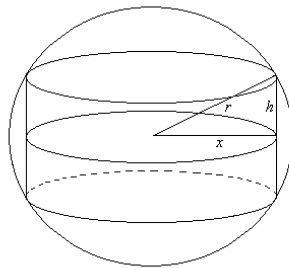
Example 1 : Prove that the volume of the sphere with radius r is $\frac{4}{3} \pi r^3$.

Solution :



$$V = 2 \int_0^r \pi r'^2 dx = 2 \int_0^r \pi (r^2 - x^2) dx$$

$$= \frac{4}{3} \pi r^3 \dots\dots\dots (\S 6.2)$$

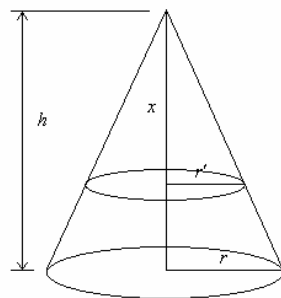


$$V = 2 \int_0^r 2\pi x h dx = \int_0^r 2\pi x \sqrt{r^2 - x^2} dx$$

$$= -\frac{4\pi}{3} (r^2 - x^2)^{3/2} \Big|_0^r = \frac{4}{3} \pi r^3 \dots\dots\dots (\S 6.3)$$

Example 2 : Find the volume of the cone with base radius r and the height h .

Solution :



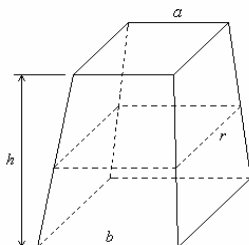
$$\frac{x}{h} = \frac{r'}{r}$$

$$\Rightarrow r' = \frac{r}{h} x.$$

$$V = \int_0^h \pi r'^2 dx = \int_0^h \frac{\pi r^2}{h^2} x^2 dx$$

$$= \frac{1}{3} \pi r^2 h.$$

Example 3: A frustum of a pyramid with square base of side b , square top of side a , and height h .



Solution :

$$V = \int_0^h r^2 dx$$

$$\frac{k}{h} = \frac{a}{b-a} \Rightarrow k = \frac{ah}{b-a}$$

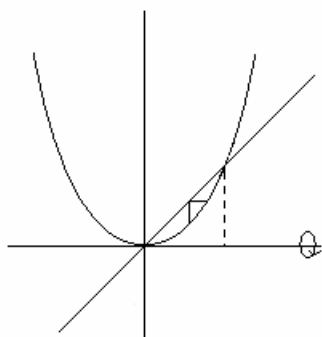
$$\frac{x+k}{k} = \frac{r}{a} \Rightarrow r = a + \left(\frac{b-a}{h}\right)x.$$

$$\begin{aligned} \Rightarrow V &= \int_0^h \left(a + \frac{b-a}{h}x \right)^2 dx \\ &= \frac{h}{3(b-a)} \left(a + \frac{b-a}{h}x \right)^3 \Big|_0^h = \frac{h}{3} (a^2 + ab + b^2). \end{aligned}$$

Example 4 : $\begin{cases} y=x \\ y=x^2 \end{cases}$ ——— x -axis

(由此 2 曲線所圍區域繞著 x 軸所旋轉出來的體積)

Solution :

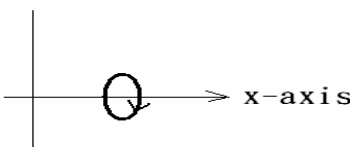


方法 1 (截面為環狀)

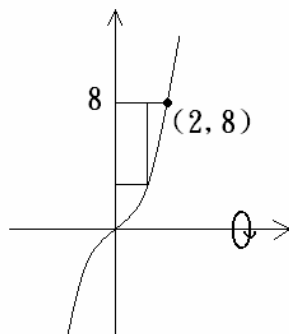
$$\begin{aligned} V &= \int_0^1 \pi r_1^2 - \pi r_2^2 dx \quad r_1 = x, r_2 = x^2 \\ &= \int_0^1 \pi x^2 - \pi x^4 dx \\ &= \frac{2\pi}{15}. \end{aligned}$$

方法 2 (截面為圓柱狀)

$$\begin{aligned} V &= \int_0^1 2\pi r h dy \\ &= \int_0^1 2\pi y (\sqrt{y} - y) dy \\ &= \frac{2\pi}{15}. \end{aligned}$$

Example 5 : $\begin{cases} y = x^3 \\ y = 8 \\ x = 0 \end{cases}$ 

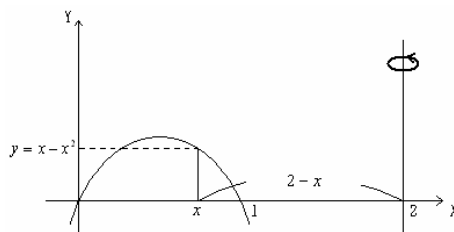
Solution :



$$\begin{aligned} V &= \int_0^2 (61 - x^6) \pi dx \quad (\text{截面為環狀}) \\ &= \int_0^8 2\pi y^{\frac{1}{3}} y dy \quad (\text{截面為圓柱狀}) \\ &= \frac{768}{7} \pi. \end{aligned}$$

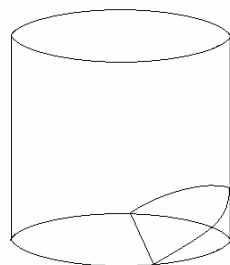
Example 6 : $\begin{cases} y = x - x^2 \\ y = 0 \end{cases} \quad x = 2$ 

Solution :



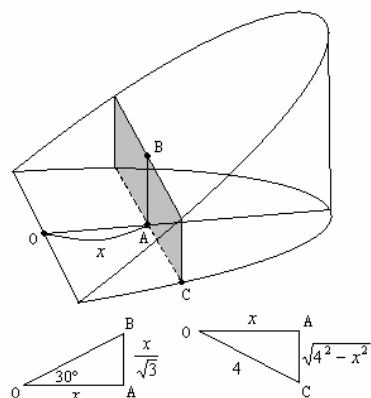
$$\begin{aligned} V &= \int_0^1 2\pi (2-x)(x-x^2) dx \\ &= \frac{\pi}{2}. \end{aligned}$$

Example 7: A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder. The other intersects the first at an angle of 30° along a diameter of the cylinder. Find the volume of the wedge.



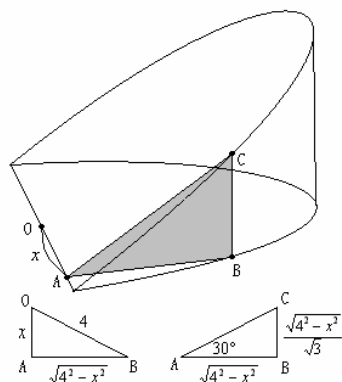
Solution :

(Method 1)



$$\begin{aligned} V &= \int_0^4 (\text{長方形截面}) dx \\ &= \int_0^4 \underbrace{2\sqrt{4^2 - x^2}}_{\text{長}} \underbrace{\left(\frac{x}{\sqrt{3}}\right)}_{\text{寬}} dx \\ &= \frac{128}{3\sqrt{3}}. \end{aligned}$$

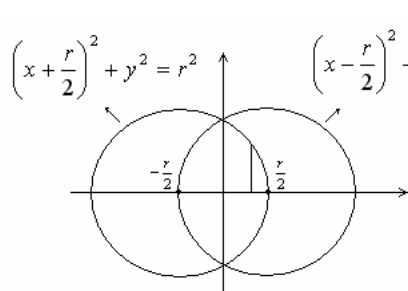
(Method 2)



$$\begin{aligned} V &= 2 \int_0^4 (\triangle ABC \text{ 截面}) dx \\ &= \int_0^4 (AB \times BC) dx \\ &= \int_0^4 \frac{16 - x^2}{\sqrt{3}} dx \\ &= \frac{128}{3\sqrt{3}}. \end{aligned}$$

Example 8 : Find the volume common to two spheres, each with radius r , if the center of each sphere lies on the surface of other .

Solution :



$$\begin{aligned} V &= 2 \int_{-\frac{r}{2}}^{\frac{r}{2}} \pi y^2 dx \\ &= 2 \int_{-\frac{r}{2}}^{\frac{r}{2}} \pi \left(r^2 - \left(x + \frac{r}{2} \right)^2 \right) dx \\ &= \frac{5}{24} \pi r^3. \end{aligned}$$

OR

$$\begin{aligned} V &= 2 \int_{-\frac{r}{2}}^{\frac{r}{2}} \pi y^2 dx \\ &= 2 \int_{-\frac{r}{2}}^{\frac{r}{2}} \pi \left(r^2 - \left(x - \frac{r}{2} \right)^2 \right) dx \\ &= \frac{5}{24} \pi r^3. \end{aligned}$$